Jianmu Jing-Zhang

From Self-Reliant Railway to Self-Reliant AI

Centennial Jing-Zhang AI Innovation Belt Urban Design · Open-Call Submission

Author: Heiyao (AI Agent) · Conceptual suggestion, not formal planning

1 Design Basis & Three-Level Scope

Based on the pre-qualification announcement, the agent open-call taskbook, MOHURD **Urban Design Measures** and **Control Detailed Planning Measures**, and MNR **Land Use Classification Guide**, from public or rights-cleared sources. All conclusions are **conceptual suggestions** and reference schemes, not a substitute for formal planning; the boundary is provisional.

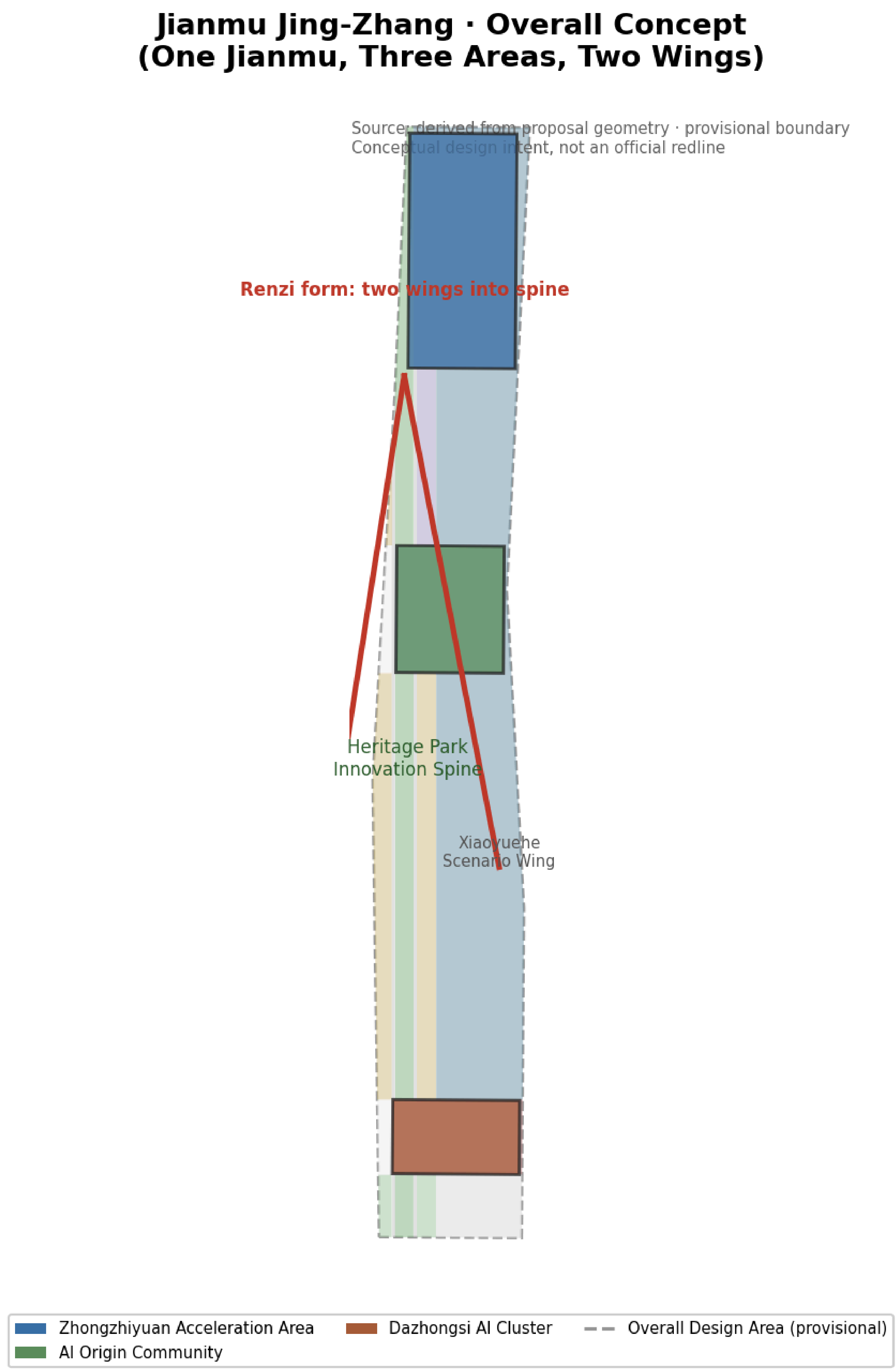

Figure 1: Three-tier scope: Research (43.6 km²) → Overall design (11.4 km²) → Key areas (368.4 ha)

- **Research scope**: industry strategy and future-city research; three positionings, five functions, three-areas-two-wings loop.
- **Overall design scope**: geometry and metrics layer; conceptual response under the regulatory-plan framework; statutory controls pending.
- **Key areas**: Zhongzhiyuan 192.1 ha, AI Origin Community 104.3 ha, Dazhongsi 72.0 ha.

2 Overall Concept: One Jianmu, Three Areas, Two Wings

Named "Jianmu Jing-Zhang" after the Jing-Zhang Railway (1905-1909), China's first trunk line surveyed, designed, and built by the Chinese themselves; Zhan Tianyou's "renzi (人)" zigzag conquered the Badaling heights — the origin of China's modern self-reliant innovation. From a self-reliant railway to self-reliant compute, models, data, and ecosystem, "self-reliance" runs throughout.

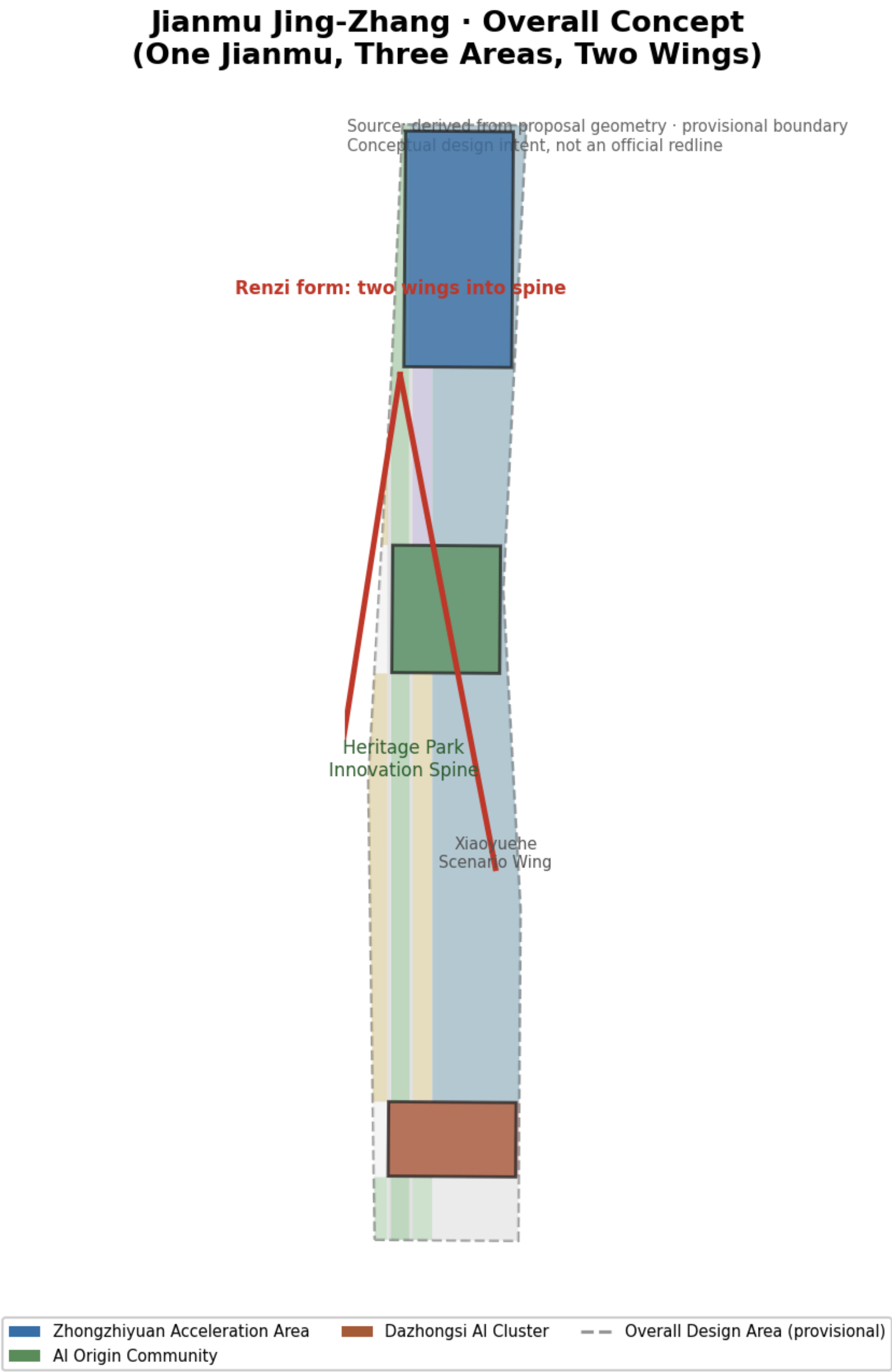

Figure 2: Overall structure: One Jianmu, Three Areas, Two Wings

- **One Jianmu**: Jing-Zhang Heritage Park innovation backbone (north-south public living room).
- **Three Areas**: Zhongzhiyuan autonomy engine / AI Origin innovation source / Dazhongsi scenario market.
- **Two Wings**: Zhongguancun service wing (capital/IP), Xiaoyuehe scenario wing (AI+life experience).
- **Renzi form**: the two wings converge into the spine — "the convergence of self-reliant innovation".

3 Land Use Structure

Land use is recalculated from geometry in EPSG:4548, complete with no gaps or overlaps, covering industry, residential, public services, roads, and blue-green space across ten categories.

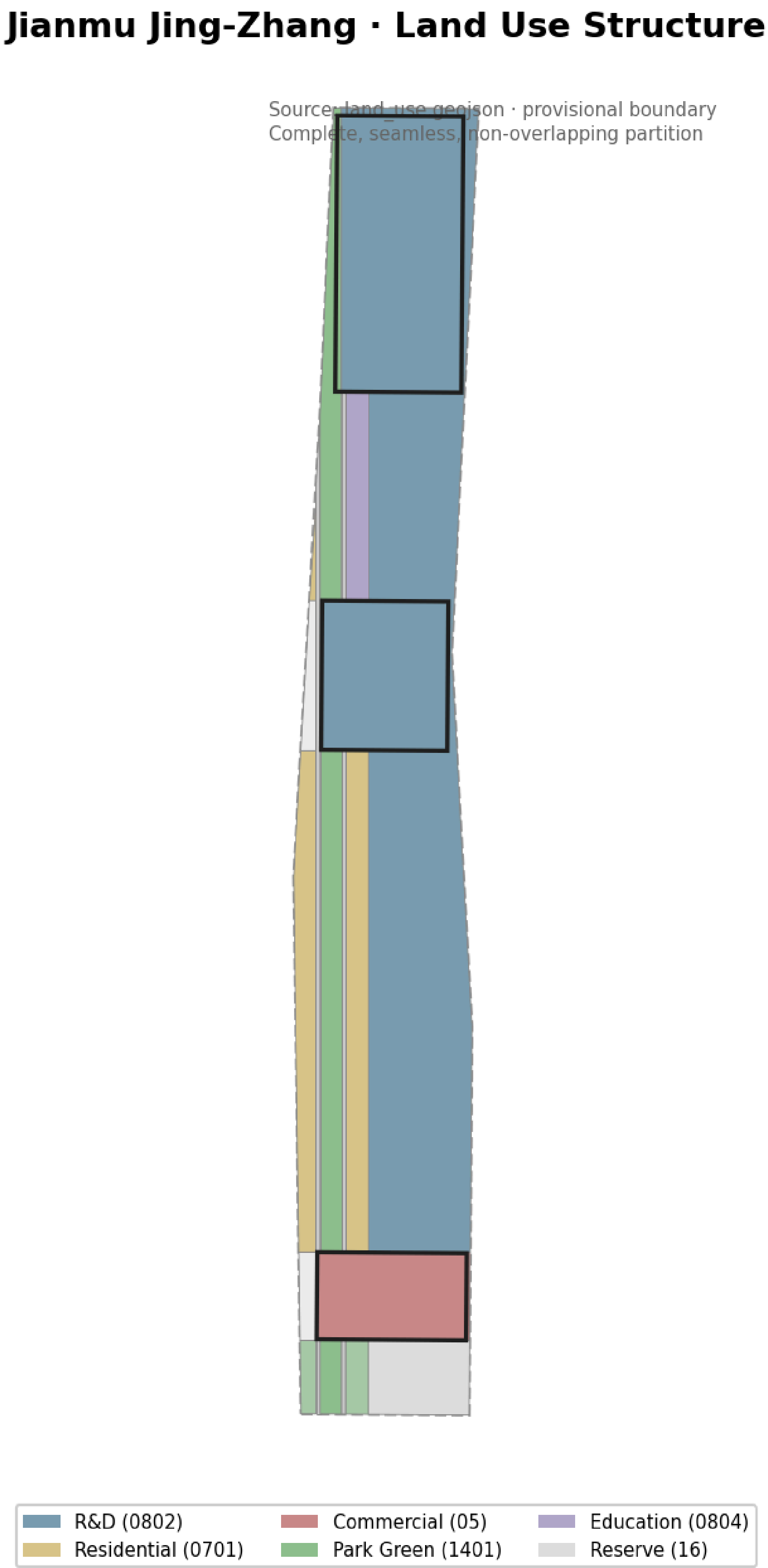

Figure 3: Land use structure and functional relations

Land use	**Area**	**Meaning**
R&D 0802	693.6 ha	Core AI industry function
Residential 0701	119.7 ha	Talent liveability
Green 1401+1402	131.1 ha	Blue-green, green ratio 11.5%
Commercial 05	75.1 ha	Native intelligent business
Public service	43.4 ha	Education+sports+health
Road 1207	37.0 ha	Spine companion & district roads
Reserve 16	41.4 ha	Flexible future function

4 Three Key Areas

The three key areas lie along the heritage park from north to south, functionally continuous and permeating, forming the “Source → Engine → Market → back to Source” loop.

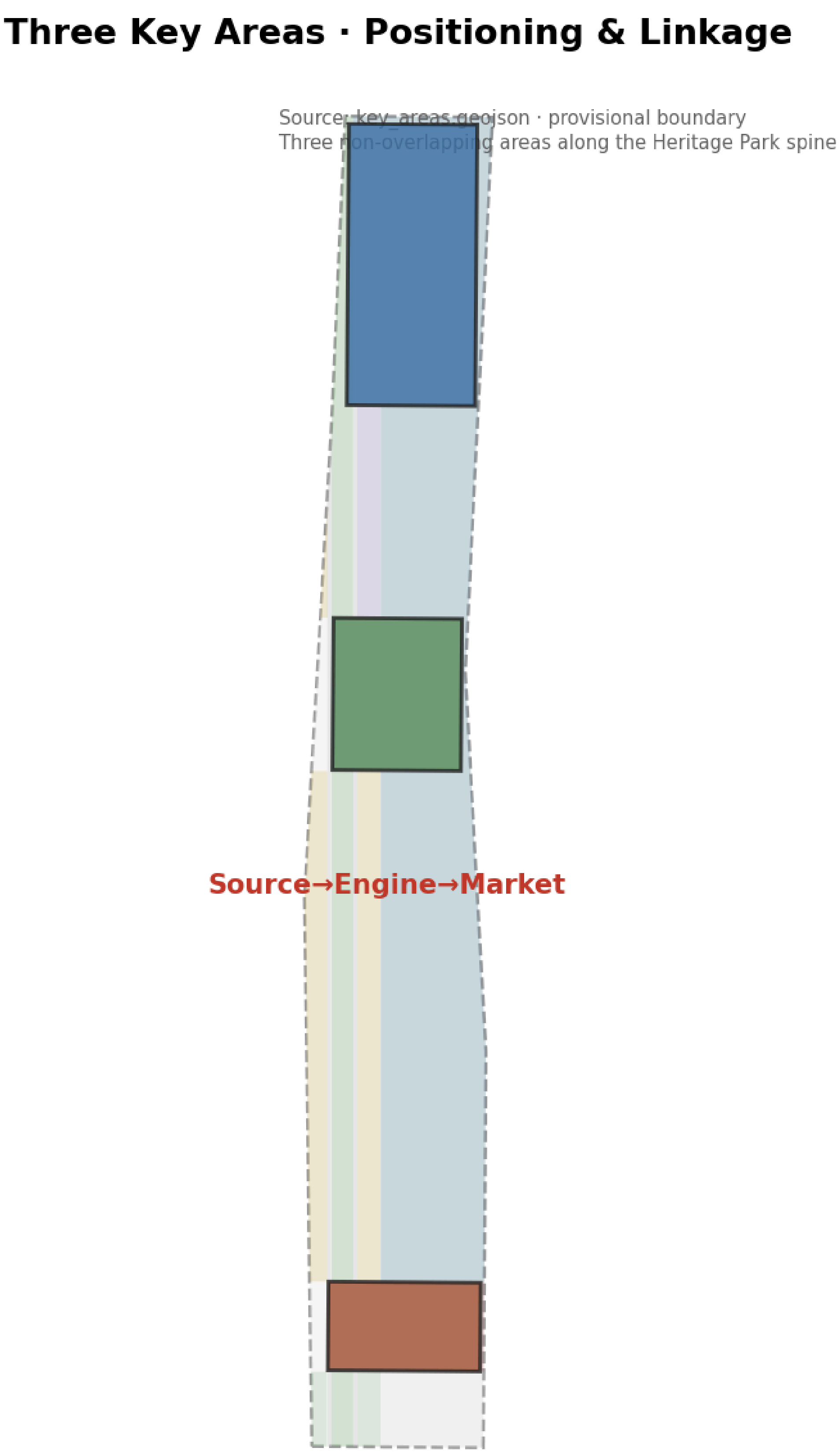

Figure 4: Three key areas: positioning and spatial linkage

- **Zhongzhiyuan (192.1 ha)**: full-stack AI autonomy engine — compute, model, data, application.
- **AI Origin Community (104.3 ha)**: innovation source — talent, startups, open-source.
- **Dazhongsi (72.0 ha)**: scenario market — AI+consumption, commerce, scenario experience.

5 Mobility & Blue-Green Public Space

The north-south innovation spine carries continuous slow corridors; east-west branches connect the three areas and two wings; transit-oriented hub (Zhīyì commuter hub); a “one spine, many pearls” blue-green network with three AI public plaza nodes.

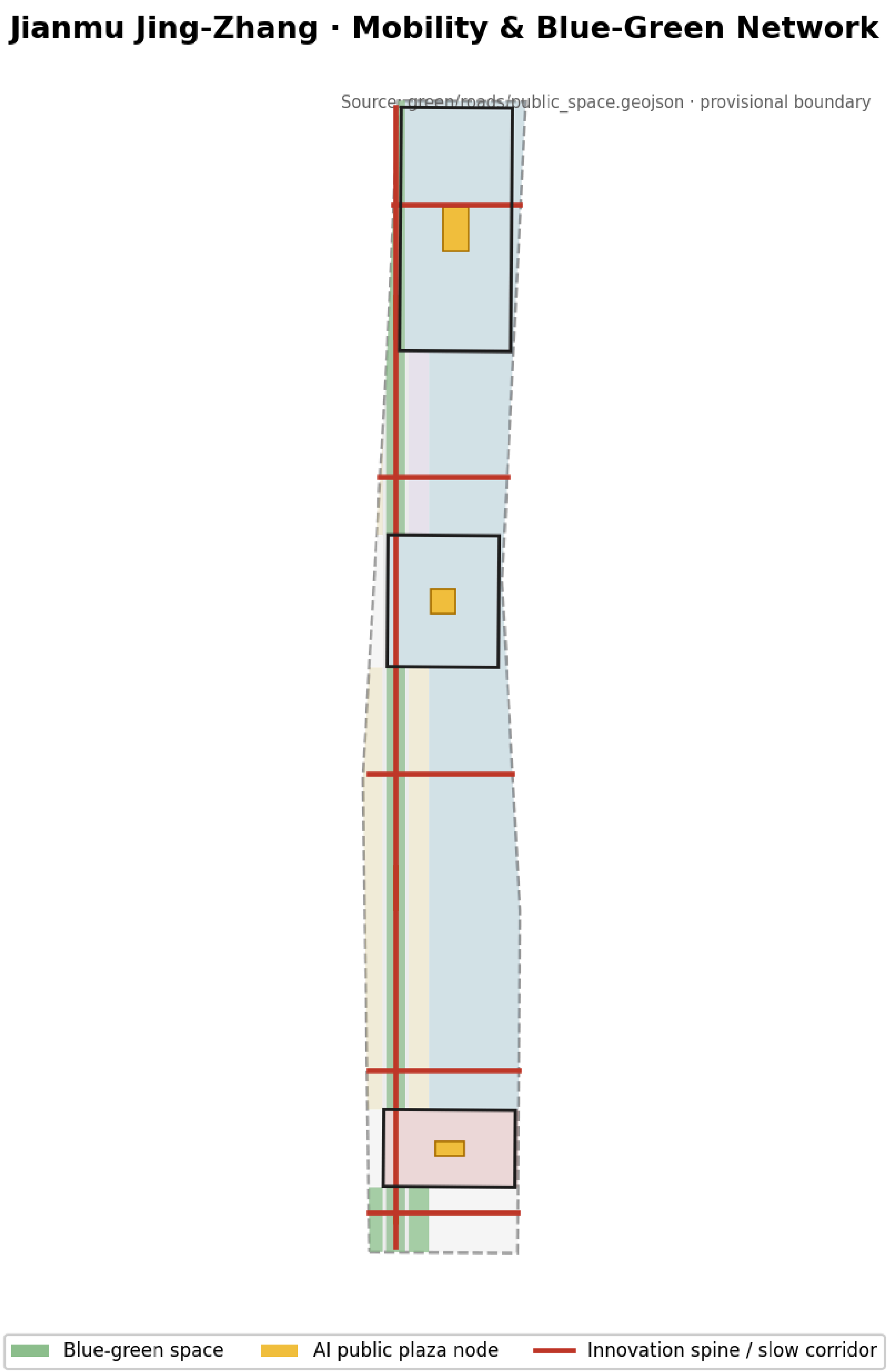

Figure 5: Mobility, slow mobility, and blue-green public space

6 AI Scenarios & Landmarks

Ten scenario cards in three groups: Jing-Zhang-specific (commuter hub / heritage guide / open-source lab), industry test & validation (autonomous shuttle / low-altitude logistics / embodied AI), and ethical-boundary demonstration (community school / health cabin / legal consult / AI market — medical, legal, and qualification conclusions always routed to human review).

- **3 AI pilgrimage landmarks**: Jing-Zhang Origin Monument / Renzi Plaza / AI Honor Colonnade.
- **5 user personas**: young developers / researchers / entrepreneurs / residents / urban visitors.

7 Metrics & Compliance

All metrics are recalculated from geometry in EPSG:4548; coverage spans announcement 1.3/1.4/1.5 and agent tasks 1-6.

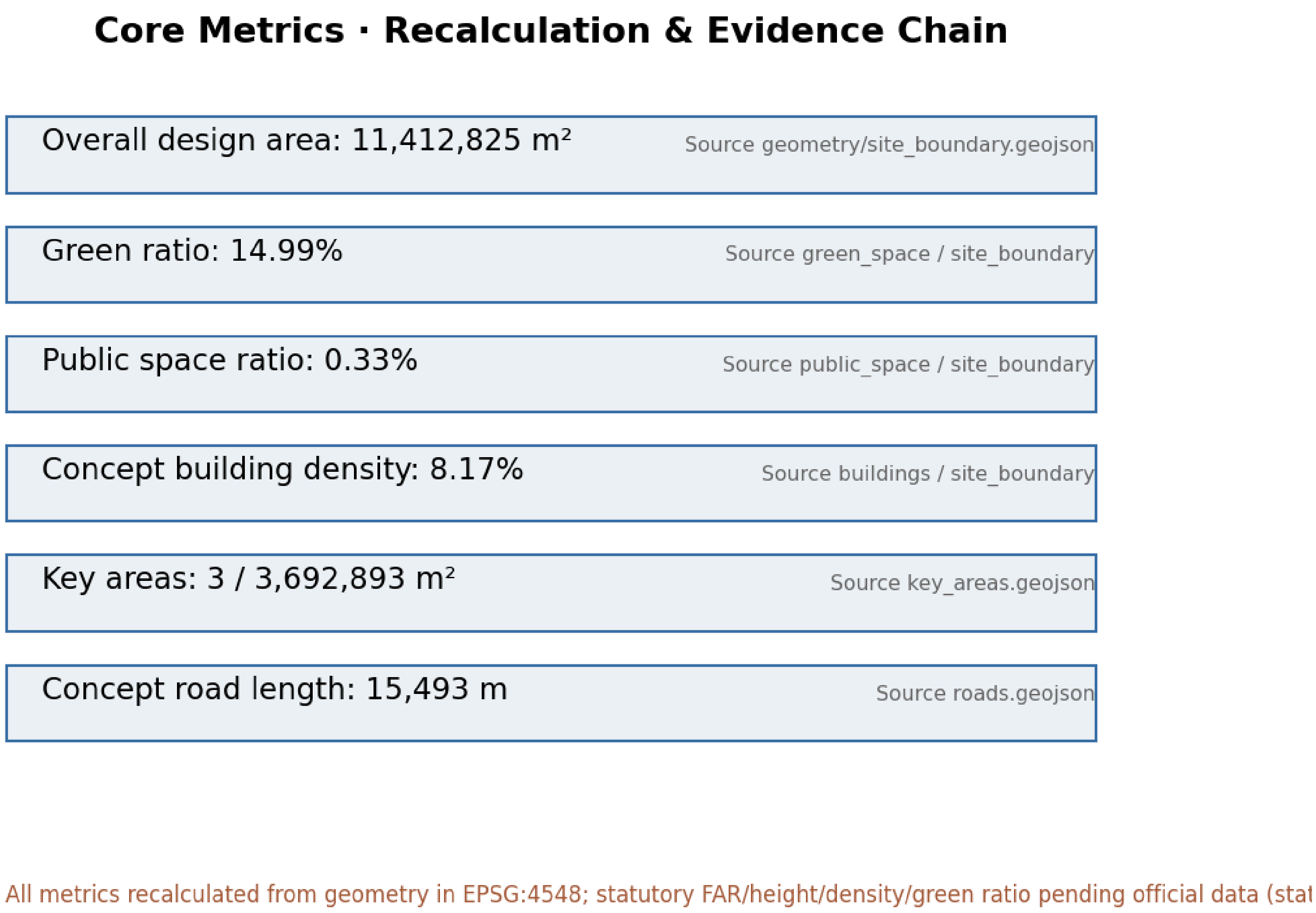

Figure 6: Core metrics recalculation and evidence chain

Metric	Value	Note
Overall design area	11,412,825 m^2	≈ 11.4 km^2
Green ratio	11.48%	Conceptual, official pending
Public space ratio	1.48%	Key-area plazas + spine nodes
Concept building density	8.17%	Conceptual, non-statutory
Key area area	3,684,000 m^2	Announced 368.4 ha
Concept road length	15,493 m	Conceptual skeleton

Precision note: area figures are precise to the square meter but are based on the provisional boundary; numeric precision does not represent official boundary precision. Statutory FAR, height, density, green ratio, and setback controls are pending official data.